Heba Qasem

Computer Systems Engineering

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EDUCATION

B.Sc of Computer Systems Engineering 2021
Palestine Technical University - Kadoorie (PTUK). Tulkarm
Average 86.7%.

Tawjihi (Scientific Stream) 2020 – 2021
Jalqamous Secondary School. Jenin
Average 98.4%.

PROFESSIONAL EXPERIENCE

Data Science & Machine Learning Intern 01/2026 – 04/2026
IBT Learning Solutions × GGateway Remote
• Completed an intensive 12-week hands-on bootcamp covering Python for Data Science, SQL, Data Exploration, Mathematical Foundations, Machine Learning I & II (tree-based models), Time Series Analysis, Neural Networks, CNNs, RNNs, NLP, Large Language Models, Recommendation Systems and Reinforcement Learning.
• Worked with industry tools: Python, Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, TensorFlow, Matplotlib, Seaborn, Google Colab, SQL, and Excel.

Backend Development Intern 10/2025 – 12/2025
Exalt Technologies Nablus
• Backend development with Java & Spring Boot: RESTful API design, dependency injection, layered architecture, and domain-driven feature-based project structure.
• Security & Auth: JWT token lifecycle, refresh tokens, multi-device session management, and role-based access control (RBAC).
• Database & integration: MySQL schema design, JPA/Hibernate ORM and cursor-based pagination

Problem Solving Internship At PTUK
145 Problem Solved on Codefoces | 81 Problem Solved on Leetcode

PROJECTS

Early Warning System for High School Dropout Risk (Capstone – IBT × GGateway)
• Built a supervised ML model to predict student dropout risk using the HSLS:09 national longitudinal dataset (~15,900 students across 50 U.S. states).
• Applied class weighting, threshold tuning, and SHAP analysis to push recall from 14% → 87% (Tuned XGBoost at threshold 0.30), catching ~9 out of 10 at-risk students.
• Compared 17 model configurations (Logistic Regression, Random Forest, XGBoost, LightGBM) and conducted ablation study showing class weighting was the dominant lever (+59 pp recall).

Skill Sharing & Mentorship Platform API (Capstone – Exalt Technologies)
• Production-grade backend for an organizational mentorship platform: posts & feeds, live session scheduling, automated mentor badge evaluation, in-app notifications, and admin moderation.
• Role-based access, event-driven side effects, multi-device

SKILLS

Languages: Python, Java, C++, SQL
ML / Data Science: Scikit-learn, XGBoost, LightGBM, TensorFlow, SHAP, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, Google Colab
ML Topics: Supervised & Unsupervised Learning, Neural Networks (CNN/RNN), NLP, LLMs, Feature Engineering, Time Series, Recommendation Systems, Reinforcement Learning
Backend & Databases: Spring Boot, JWT, REST APIs, MySQL, SQL Server, Maven, Postman, Git/GitHub

LANGUAGES

English (Professional) Arabic (Native)